

Compressed Regret Sensing: Environment Design without Reproducing the Data-Generating Process

Haoyang Shang (UBC) · Xuan Liu (UC San Diego)

Course project proposal, Reinforcement Learning (UBC). Companion to the Machine Learning course proposal; both form one paper.

1. Motivation

Training self-improving, long-horizon agents with RL needs either scarce real interaction data or full rollouts in a simulator that reproduces the whole data-generating process, which is wasteful because most steps of a long episode are routine. Unsupervised environment design (UED) avoids the data problem by generating training environments: PAIRED [1] trains an adversary to maximize the agent's *regret*, and PLR/ACCEL [2,3] replay and edit high-regret levels. All of them estimate regret *one environment at a time with full rollouts*, and PAIRED's generator gets one sparse reward per environment. We ask two questions: **how few rollouts are needed to know where an agent is deficient, and which parts of an episode need to be rolled out at all?**

2. Setting

An underspecified POMDP $\mathcal{M} = \langle A, O, \Theta, S, T, I, R, \gamma \rangle$ yields a POMDP $\mathcal{M}_\theta$ for each free parameter $\theta \in \Theta$. Let $V(\pi, \theta)$ be the expected discounted return and $\text{Reg}(\pi, \theta) = V(\pi_\theta^*, \theta) - V(\pi, \theta)$. PAIRED approximates π_θ^* with an antagonist; at Nash equilibrium the protagonist is minimax-regret. We assume access only to a small evaluation set $\mathcal{E} = \{\tilde{\theta}_1, \dots, \tilde{\theta}_m\} \sim P^*$ with a verifier, and *no* generator of P^* .

3. Structural assumptions: two kinds of sparsity

(A1) Skill sparsity. There exist a skill-demand map $\phi : \Theta \rightarrow \mathbb{R}^d$ and a deficiency vector $\delta(\pi) \in \mathbb{R}^d$ with

$$\text{Reg}(\pi, \theta) = \phi(\theta)^\top \delta(\pi) + \varepsilon, \quad \|\delta(\pi)\|_0 \leq k \ll d.$$

Equivalently, the regret matrix $R_{ij} = \text{Reg}(\pi_i, \theta_j)$ over a population of agents and environments has rank $\leq d$.

(A2) Temporal sparsity. In an H -step episode, $|\{t : |A^\pi(s_t, a_t)| > \epsilon\}| \leq s \ll H$: only a few decision-critical steps carry the learning signal.

4. Method

(a) Randomize to measure. Pick n probe environments and measure $y_i = \hat{V}(\pi_{\text{ref}}, \theta_i) - \hat{V}(\pi, \theta_i)$, where π_{ref} is the antagonist or the best agent in the population. Recover the deficiency by basis pursuit:

$$\hat{\delta} = \arg \min_{\delta} \|\delta\|_1 \text{ s.t. } \|y - \Phi \delta\|_2 \leq \eta, \quad \Phi = [\phi(\theta_1), \dots, \phi(\theta_n)]^\top.$$

If Φ satisfies the RIP of order $2k$ with $\delta_{2k} < \sqrt{2} - 1$, then $\|\hat{\delta} - \delta\|_2 \leq C_0 k^{-1/2} \|\delta - \delta_k\|_1 + C_1 \eta$ [4], and for sub-Gaussian designs $n = O(k \log(d/k))$ probes suffice. *Domain randomization is a poor training distribution but a good sensing design*; we use it only to measure.

(b) Learn ϕ by regret-matrix completion. ϕ is unknown for rich Θ . A population of agents gives sparse entries Ω of R ; we solve $\min_X \|X\|_*$ s.t. $P_\Omega(X) = P_\Omega(\hat{R})$ (or factorized ALS), which needs $|\Omega| = O(r(N+M) \text{ polylog})$ entries instead of NM [5]. A regressor $g_\omega(\theta) \approx \phi(\theta)$ extends it to unseen θ .

(c) Target to train. The generator G_ψ , an RL policy that writes θ step by step as in PAIRED, receives a dense predicted-regret reward with a solvability term:

$$r_G(\theta) = g_\omega(\theta)^\top \hat{\delta} - \lambda \mathbf{1}[\hat{V}(\pi_{\text{ref}}, \theta) < \tau].$$

This replaces PAIRED's sparse reward and removes per-environment antagonist rollouts; $\hat{\delta}$ is re-sensed every K updates.

(d) Sparse rollouts. Flag critical steps by ensemble value disagreement or policy entropy, run routine segments with cached macro-actions (options) [6], and branch Monte-Carlo rollouts only at flagged states: cost $O(sKL)$ instead of $O(NH)$ for K branches of length L .

5. Experiments

Testbeds JAX minimax library: MiniGrid mazes (the PAIRED benchmark) with zero-shot transfer to Labyrinth, FourRooms and SixteenRooms. For long horizons, Craftax.

Baselines Domain randomization, minimax adversary, PAIRED, Robust PLR, ACCEL.

Metrics Zero-shot solve rate vs. environment steps; error in $\hat{\delta}$ vs. number of probes n ; singular spectrum of $\hat{R}$; fraction of steps flagged critical and compute saved.

H1 (tests A1): the top k singular values of $\hat{R}$ capture most of its energy. **H2**: CRS matches PAIRED/PLR transfer with substantially fewer environment steps. **H3** (tests A2): sparse rollouts reduce simulation cost on Craftax without lowering return. **Risk**: if R is not low-rank, we report the spectrum as a finding and fall back to PLR-style replay prioritized by predicted regret.

6. Plan

Wk 1–3 reproduce PAIRED/PLR in minimax; **4–6** probes, matrix completion, test A1; **7–10** CRS generator, sparse rollouts; **11–13** transfer experiments, write-up.

Deliverables: code, report, method section of the joint paper.

[1] Dennis et al. Emergent Complexity and Zero-shot Transfer via Unsupervised Environment Design (PAIRED). NeurIPS 2020. [2] Jiang et al. Replay-Guided Adversarial Environment Design (Robust PLR). NeurIPS 2021. [3] Parker-Holder et al. Evolving Curricula with Regret-Based Environment Design (ACCEL). ICML 2022. [4] Candès. The restricted isometry property and its implications for compressed sensing. C. R. Math. 2008. [5] Candès, Recht. Exact matrix completion via convex optimization. FoCM 2009. [6] Sutton, Precup, Singh. Between MDPs and semi-MDPs (options). AIJ 1999.